

JEE MAIN REVISION MASTER: CHEMISTRY IN EVERYDAY LIFE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Therapeutic Drug Classes: Analgesics, Antibiotics & Antiseptics	8 PYQs	Pharmaceutical Chemistry
Moderate	Food Additives, Artificial Sweeteners & Detergents	3 PYQs	Household Chemistry

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Therapeutic Drug Classes: Analgesics, Antibiotics & Antiseptics	$\text{Aspirin: 2-acetoxybenzoic acid}$, $\text{Dettol: Chloroxylenol} + \alpha\text{-Terpineol}$
Food Additives, Artificial Sweeteners & Detergents	$\text{Cationic: } [\text{CH}_3(\text{CH}_2)_{15}\text{N}(\text{CH}_3)_3]^+ \text{Br}^-$, $\text{Anionic: } [\text{CH}_3(\text{CH}_2)_{11}\text{OSO}_3^- \text{Na}^+]$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Therapeutic Drug Classes: Analgesics, Antibiotics & Antiseptics (High Yield — 8 PYQs)

Core Principle: Analgesics relieve pain: non-narcotics (aspirin, paracetamol) inhibit prostaglandin synthesis; narcotics (morphine, codeine) are habit-forming pain-relievers. Antiseptics: Dettol (chloroxylenol + terpineol), bithionol (added to soaps), tincture of iodine (2-3% iodine in alcohol-water). Disinfectants: 1% phenol solution (0.2% phenol is antiseptic).

■ **Caution / JEE Pitfall:** Phenol acts as an antiseptic at 0.2% concentration, but acts as a disinfectant at 1.0% concentration!

■ **Speed Trick:** Bactericidal antibiotics kill bacteria (penicillin, ofloxacin); bacteriostatic antibiotics inhibit bacterial growth (erythromycin, tetracycline).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Match the compounds of List-I with their effects in List-II: List-I List-II (a) Chloramphenicol (i) Malaria (b) Thyroxine (ii) Anaesthetic (...)" [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "Which one of the following gases converts haemoglobin into carboxy haemoglobin?..." [Ans: N/A]
- [2023 Shift] Q#4 (MCQ): "Which one of the following is a non-narcotic analgesic?..." [Ans: N/A]

3.2 Food Additives, Artificial Sweeteners & Detergents (Moderate Yield — 3 PYQs)

Core Principle: Artificial sweeteners: Aspartame (unstable at cooking temperatures), Saccharin (550x sweeter), Sucralose (stable at cooking temperatures), Alitame (high potency, 2000x sweeter). Detergents: Anionic (sodium lauryl sulphate), Cationic (cetyltrimethylammonium bromide, germicidal), Non-ionic (polyethylene glycol stearate).

■ **Caution / JEE Pitfall:** Aspartame cannot be used in baking or cooked foods because it decomposes at elevated temperatures.

■ **Speed Trick:** Cationic detergents have germicidal properties and are used in hair conditioners and hospital sanitation.

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): "The compound is..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Which one of the following is a cationic detergent?..." [Ans: N/A]
- [2020 Shift] Q#4 (MCQ): "A food additive that acts as an antioxidant is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Match the compounds of List-I with their effects in List-II: List-I List-II (a) Chloramphenicol (i) Malaria (b) Thyroxine (ii) Anaesthetic (c) Chloroquine (iii) Goiter (d) Chloroform (iv) Typhoid fever

- (A) a-i, b-ii, c-iii, d-iv
- (B) a-iv, b-iii, c-i, d-ii
- (C) a-i, b-iii, c-iv, d-ii
- (D) a-iv, b-iii, c-ii, d-i

Correct Answer: None

Solution Analysis: Chloramphenicol is an antibiotic. It is used in the treatment of typhoid fever. Thyroxine is a thyroid hormone. Its deficiency is related to goiter, so it matches with goiter. Chloroquine is an antimalarial drug, so it is used for malaria. Chloroform has been used as an anaesthetic. So the correct matching is: (a) → (iv), (b) → (iii), (c) → (i), (d) → (ii). Hence, the correct option is: Option B

Benchmark Problem 2: KCET 2024 (Q#2)

The compound is

- (A) sucralose
- (B) aspartame
- (C) saccharin
- (D) alitame

Correct Answer: None

Solution Analysis: The given structure is a phthalimide but no option of phthalimide is given. However, among the given options it is similar to saccharin. The structure of saccharin is